



Sound generation by water drop impact on surfaces

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ABSTRACT

In this work, we experimentally study the acoustic response of water drop impacts onto a variety of surfaces using both audio recording and high-speed imaging. Audio recordings of the airborne sound emitted by single drop impacts on deep pools, thin liquid films, and dry aluminum and masonry surfaces are captured in the laboratory. Longer audio recordings are also collected for natural rainfall on wet concrete surfaces. These airborne sound pressure versus time measurements are converted to power spectra using the fast Fourier transform. Each impact type results in unique features in the spectra from 1 to 20 kHz that can be related to the impact and subsequent fluid phenomena. For example, in the case of water drops impacting a deep pool of water, we observe two distinct sound events: one as the drop breaks through the free surface of the pool with frequency content centered around 10 kHz and then approximately 50 ms later at the time of air cavity collapse and Worthington jet formation with frequency content around 4 kHz. A Helmholtz resonator model for the collapse of an air cavity describes the observed acoustic signature well. The acoustic spectra of various natural rainfall events on wet surfaces are comparable to those of discrete drops recorded in the laboratory, but upshifted in frequency.

1. Introduction

The impact of a liquid drop on a solid or liquid surface is a complex fluid dynamics phenomenon that occurs in natural and engineered systems. As high-speed imaging technology has improved, almost every stage of drop impacts has been scrutinized in laboratory environments designed to parallel theory or emulate the conditions found in rainfall, manufacturing, or industrial processes. Observations of the moments before, during, and after drop collision show the evolution of multiple fluid dynamics phenomena involving exchanges between inertial, aerodynamic, viscous, and surface tension forces [1–5].

Whether through scientific observations or in everyday experiences, most people can recognize the liquid drop impact nature of rainfall and its effect on their world. Precipitation measurements are used to monitor foliar health, estimate soil erosion, and predict water levels of rivers, lakes, and reservoirs. Many have observed the convenience of a timely rainstorm on a parched landscape or contrarily the annoyance of a loud storm during the night. Other instances of drop ejection and impact include the expulsion of contaminated liquid during human or animal coughing that contribute to disease transmission. Additionally, drop impacts are important in a variety of technological applications including ink-jet printing, liquid fuel (spray) combustion, fire

suppression, and sonar.

Here we investigate the airborne acoustic signature of single drop impacts onto liquid or solid surfaces in the laboratory. These laboratory experiments provide a controlled environment for direct sound pressure amplitude measurements versus time, which can be post-processed into the spectral domain. Simultaneous high-speed imagery allows a connection between those sounds generated by drop impact events and the observed fluid dynamics. We also take outdoor measurements of the sounds resulting from natural rainfall impact on wet surfaces for comparison to the laboratory experiments.

The fluid response of a liquid drop impact onto a substrate depends on a number of fluid and surface properties. Generally, the Weber number and K-numbers are used to parameterize drop impact events. The Weber number, relates the tendency of the drop's kinetic energy to overwhelm the constraint provided by its surface energy, and is given by $We = \frac{\rho V^2 D}{\sigma}$; where ρ is the fluid density, V drop velocity, D drop diameter, and σ surface tension. The K-number, has been proposed [6] as a modification to the Weber number to account for viscous effects on the threshold for splash or deposition in the impact of a drop on a surface, and is given by $K' = We \cdot Oh^{0.4}$; where $Oh = We^{0.5}/Re$ is the Ohnesorge number and provides dependence on the viscosity (ν) through the Reynolds number ($Re = VD/\nu$).

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As We and K' increase, a hypothetical drop will progress through a range of responses of linear, nonlinear, and then chaotic nature. Examples on a solid substrate include wetting, axisymmetric spreading and recoil, unstable lead edge spreading, and drop fragmentation. If the impacting surface is liquid, coalescence, radial jetting from the contact point, crown splash, and Worthington jet formation. The Worthington jet is a thin column of liquid that ejects at high speed vertically from the collapsing cavity formed by an impacting drop in a liquid pool. For sufficiently high impact energy, the jet rises and pinches off droplets near its maximum extent. The Weber and K numbers, however, do not account for surface chemistry, roughness, miscibility of drop and pool liquids, drop shape, and other factors [2,3,7]. In practice, an understanding of the impact response enables engineering of fluid and surface properties to produce desirable drop impact behaviors or the suppression of those deemed undesirable. Here, we seek surface properties that affect the airborne sound generation characteristics of drop impacts.

Visual studies of drop impacts began in the early days of photography with Worthington publishing single images of drop impacts in 1876 [8,9]. However, the acoustics of drop impacts remained unstudied for some period of time. To our knowledge, the earliest scientific work on the acoustics of drop impacts was reported by Franz in 1959 [10]. Franz performed a study on the underwater acoustic signature of drops, sprays, and solid objects impacting a very deep liquid pool. This study concludes that these underwater sounds recorded upon the entry of a foreign solid object are primarily comprised of the impact of the object, the object's resonant vibrations within the pool, and the excitation of entrained air bubbles. For liquid objects the sounds were categorized as an instantaneous impulse upon impact and subsequent damped oscillating waves. The initial impulse has a magnitude proportional to the drop radius and the cube of the drop velocity. No predictive theory was proposed concerning the waves, but Franz suggested the source of this sound was due to periodic vibrations of the air bubbles. Similarly to Franz's work, the majority of drop acoustic studies have focused on the below-surface acoustics of an impinging fluid into a deep liquid pool.

In 1972, Kinnell performed one of few studies on the acoustics in air for liquid drop impact onto solid surfaces [11], with primary focus on the elasticity of the drop's deformation upon impact and the amplitude of sound generated by impacts. Spectral analysis is briefly addressed, with frequencies around 300 Hz reported for water drops impacting a smooth surface and higher frequencies observed upon the addition of a thin water film on the surfaces.

Prosperetti et al. [12] and Pumphrey et al. [13] proposed that underwater sound in a rain event and in laboratory experiments has two primary sources. First, a sound resulting from the initial impact of the drop through the liquid's free surface. The second source, observed in some, but not all impacts, is driven by entrapped and excited air bubbles that generate sound. The vibrations of these bubbles create significant pressure underwater with frequencies in the range of 15 kHz. This second sound from air bubbles is considered by Pumphrey and Crum to be significantly louder than the one from initial impact. Guo and Williams formed a theory regarding the sound produced by the initial impact of a drop into a liquid pool [14]. Their analytical study affirmed the conclusion of Pumphrey and Crum regarding the acoustic energy of the drop impact being proportional to cube of the impact velocity. The frequency of the sound generated is not addressed.

Medwin et al. claimed there are three regimes of drop size that cause different acoustic responses following impact into a pool of water [15]. They characterized these regimes based on the contributions of both initial surface impact and subsequent bubble vibrations on the overall underwater sound. Of note, these authors also find that small bubble oscillations have an average recorded frequency near 15 kHz. The drops used in our laboratory studies (2.1 mm in diameter) fall into the mid-size and large drop regimes examined by Medwin et al. Their analysis concludes that these size regimes result in an initial broadband response upon impact followed by secondary sound due to oscillating bubbles at characteristic frequencies from 2 to 10 kHz.

Nystuen developed the robust acoustic rainfall analysis (ARA) method for quantifying the characteristics of rainfall from underwater sound [16]. Using data collected over 6 months, ARA was able to detect rainfall rate and drop size distribution based on underwater acoustic amplitudes and frequencies. These techniques have been further developed and detailed by Nystuen and collaborators [17].

Despite the challenging temporal and spatial scales of drop impacts compounded by the multiphase nature of the problem, computational methods have also been applied to this problem. Tajiri et al. [18] produced a simulation detailing the sound produced in air and in the bulk liquid during drop impact into a pool using a finite difference lattice Boltzmann method for two-phase flow. Their findings agree with those of Prosperetti et al. [12] and Pumphrey et al. [13], showing entrained air bubbles as the primary source of sound.

In the present work, we record airborne sounds, longitudinal compressible waves that are driven into the air around the drop impact event, not beneath the surface of a liquid pool or in the solid substrates, for four different impact cases. These recordings are converted to power spectra using Fourier transforms. These acoustic measurements are complimented by high-speed imaging of the associated impact events, with efforts to correlate sound signatures with observed fluid-surface interaction.

2. Experimental method

The experimental setup is shown in Fig. 1 and is similar to that used in prior work [19–21]. Deionized water (Evoqua deionizer) was loaded into a glass syringe with stainless steel needle. The syringe was inserted into a syringe pump (New Era NE-300) and its needle connected to a polytetrafluoroethylene (PTFE) capillary tube with a 1 mm inner diameter and 2 mm outer diameter. These materials are chemically stable to cleaning solvents including acetone, isopropanol, and methanol. The syringe pump is operated a low enough flow rate, producing drops with a diameter of 2.10 ± 0.04 mm on the tip of the capillary tube before falling under their own weight. The tip of the PTFE tube is secured 2.75 m above the impact surface and drops fall freely to impact onto the solid or liquid surface of interest with an impact velocity of 5.9 ± 0.4 m/s ($\sim 85\%$ of terminal velocity), measured via high-speed imaging. These parameters correspond to a Weber number of 1,010 and K number of 1410.

Note, that while the experiments reported here are for deionized water, no discernible differences were observed for experiments carried out using tap water or distilled water, due to the fact that the high kinetic energy at impact overwhelms any surface chemistry effects that might result from impurities or surfactants present in the water. Additionally, experiments were originally carried out for three drop diameters (1.9, 2.1, and 2.36 mm); however, results for three data sets were not appreciably different and, hence, we only present results for the 2.1 mm diameter drops here.

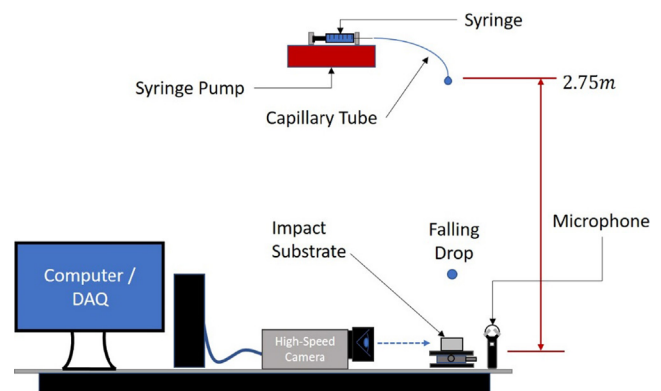


Fig. 1. Experimental setup for drop impact study.

Drops were impacted onto a variety of surfaces to examine the effects of surface roughness and pooling: dry masonry brick with surface features of a similar scale (1 mm) as the drop size, block of aluminum with fine-sanded smooth surface finish (polished to surface roughness of $R_a = \sim 25 \mu\text{m}$), smooth aluminum surface ($R_a = \sim 25 \mu\text{m}$) wetted with a thin film of water (film thickness of 4 mm), and a deep liquid pool of water. Both the masonry and aluminum substrates were substantially large ($\sim 5 \text{ kg}$), to reduce their vibrations' contribution to the recorded airborne noise. The surface features on the aluminum are extremely small relative to the drop diameter and is effectively perfectly smooth; however, the surface features on the masonry are similar in size to the drop diameter and, hence, fluid is directed into surface features causing air compression in cavities and fluid jetting which contribute to sound generation.

The impacts were backlit and imaged using a high-speed camera (MotionPro X3) and lens (AF-S Micro Nikkor 40 mm) at 400–5000 frames per second, with a maximum resolution of 1280×1024 , and an exposure time of $2.6 \mu\text{s}$. Audio recordings were acquired using a microphone (Zoom model H1N, 96 kHz at 24 bit). The microphone was nominally placed 6 cm from the impact point, just above the impact surface, to capture the airborne sound generated by the impact and subsequent fluid motion (e.g., air cavity collapse). As shown in Fig. 2, direct comparisons of sound intensity require careful microphone placement, the sound intensity for a laboratory drop impact decreased as the square of distance from impact location to microphone. This behavior is expected from the inverse square law of sound intensity based on scaling the surface area of successively larger spheres as the sound propagates.

Laboratory experiments were performed in an HVAC-controlled room at a temperature of $23 \pm 1^\circ\text{C}$, with low relative humidity ($< 20\%$), and with minimal air circulation. Outside experiments were also performed during a rainstorm to record audio of natural rainfall (moderate rainfall at $2 \pm 0.5 \text{ mm/hour}$) impact on wet concrete using the Zoom microphone. The microphone was protected from moisture by a thick layer of cloth, which absorbed the rainfall and baffled any potential sound. Locations were selected based on their isolation from traffic, structures, and other objects that could contribute spurious acoustic signatures.

3. Results

In laboratory experiments, the four substrates used as the impact

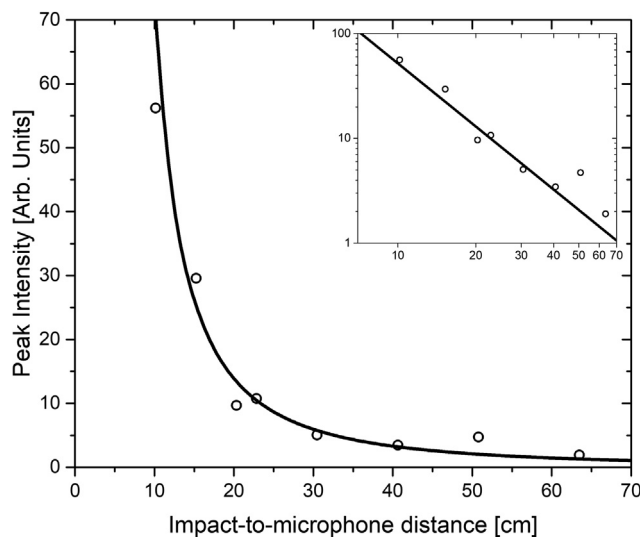


Fig. 2. Peak recorded airborne sound intensity, for drop impact onto dry masonry, as a function of distance from impact point to microphone with inverse square fit (inset is log–log scale representation of measurements).

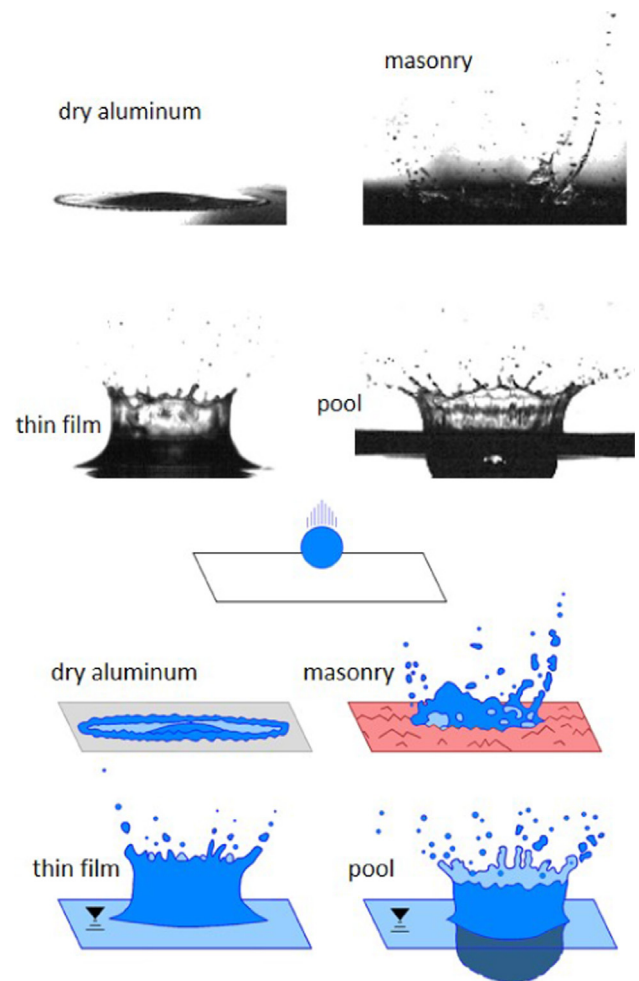


Fig. 3. Depictions of pre-impact and post-impacts onto the four tested substrates (top: photos; bottom: cartoons created from photos).

targets emulate a variety of natural or man-made surfaces. The deep pool mimics bodies of water both small and large, the dry/wet aluminum surface simulates a variety of man-made structures (e.g., car or solar panel) at various stages of a precipitation event, and the porous masonry brick emulates many construction materials used in the built environment. Fig. 3 shows images and schematics of impacts onto these surfaces and qualitatively highlight some of the observed fluid behaviors. The drop impact onto dry aluminum spreads into a thin disk with a periodic destabilized advancing edge. Impact onto the masonry brick results in a very rapid and chaotic breakup of the fluid into thin jets as the liquid is interacts with the roughness of the surface. Impacts on the liquid film and pool begin with radial jetting and evolve into a crown splash within the first few milliseconds after impact. The highly nonlinear behaviors, including jetting and droplet ejection, are expected to be significant contributions to any recorded sound.

For drop impact into a deep pool, following impact, a crown splash rises vertically and secondary droplets are formed on the destabilized rim. Images acquired for this process are detailed in Fig. 4. Of note, for both fluid and acoustic responses, is an air cavity pushed into the bulk fluid and observed within the crown splash. After expanding for $\sim 25 \text{ ms}$ following impact, the cavity recedes towards its center for approximately the same amount of time. Due to the lack of direct observation of internal air cavity with the present camera angle, quantification of the cavity size dynamics from the present photographs, such as those shown in Fig. 4, is difficult. However, measurements of the dynamics of the outer crown diameter in the present study agree well with the experimental, computational, and theoretical work of

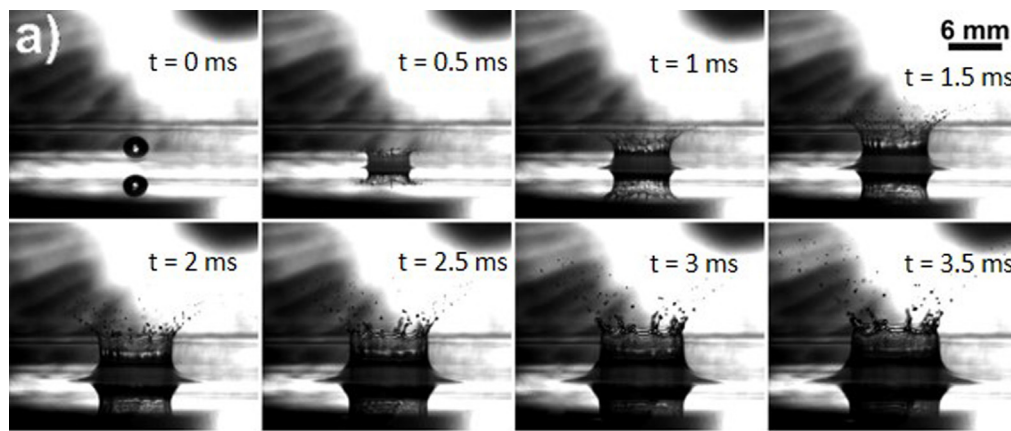


Fig. 4. (a) Photographs capturing the early stages of a water drop impact into a deep water pool ($We = \sim 1500$ and $K' = \sim 1950$). The high kinetic energy of the impacting drop pushes a crown-like sheet which destabilizes into a crown splash. (b) Illustration of crown splash formation with features labeled. Figure from Tilger [21].

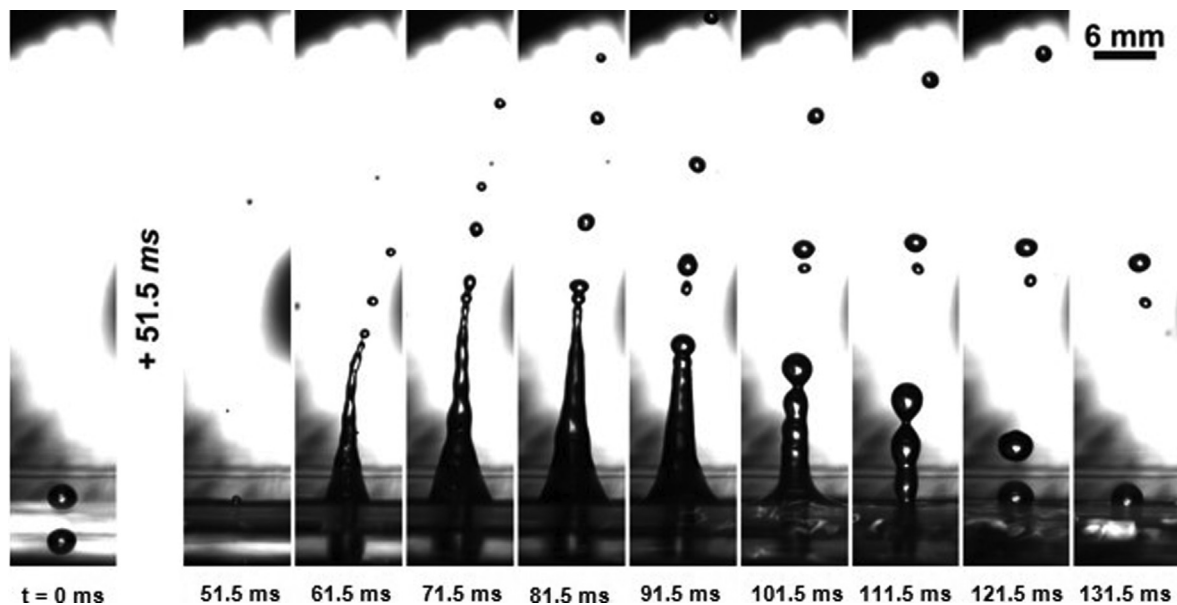
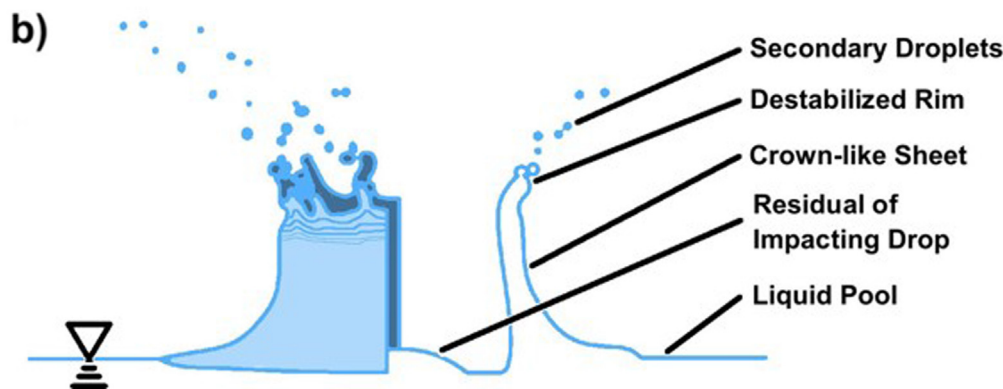


Fig. 5. Photographs capturing the late stage Worthington jet of water drop impact into a deep water pool ($We = \sim 1500$ and $K' = \sim 1950$), demonstrating the jet that issues from the impact cavity (present, but not observed in 51.5 ms photograph). Figure from Tilger [21].

Berberovic et al. [22], who have comprehensively described the air cavity dynamics for drop impact into deep pools and liquid films. The singular focusing of energy during the air cavity collapse produces a Worthington jet [23], as shown in Fig. 5 (the much later in time view of the events in Fig. 4). This jet extends out of the bulk liquid, destabilizes via the Plateau-Rayleigh instability, and ejects a train of secondary drops and/or droplets [24]. Although fewer in quantity, the largest of these fluid parcels approach the size of the impacting drop, exceeding the scale of those originating from the crown. A similar spreading and

receding occurs for impacts onto the thin liquid films. Jetting is observed in this case as well.

A raw trace of the measured sound amplitude from a single microphone channel for water drop impact into the deep pool is shown in Fig. 6. The radiated acoustic signature shows two distinct sounds that are separated by approximately 50 ms. Based on previous studies, synchronized imaging, and sound analysis: we determine the first lower intensity sound recorded at ~ 5 ms occurs when the falling drop impacts the surface of the liquid pool. The second and significantly more

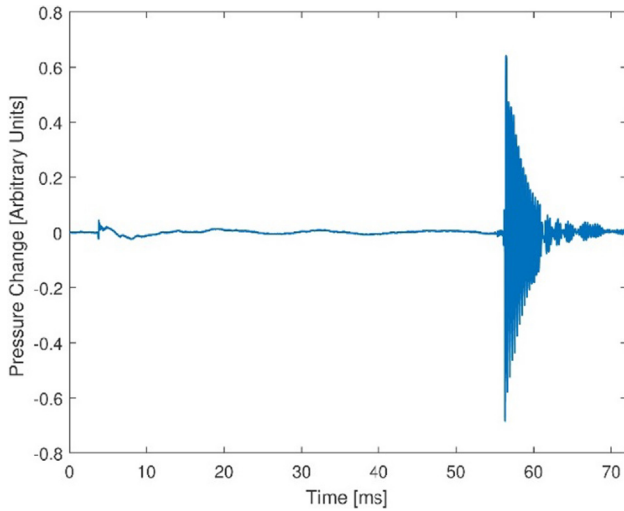


Fig. 6. Audio for drop impacts in deep pool: drop contacts free surface ~ 5 ms (2.1 mm drop at 5.9 m/s impact velocity) and cavity collapse around 55 ms.

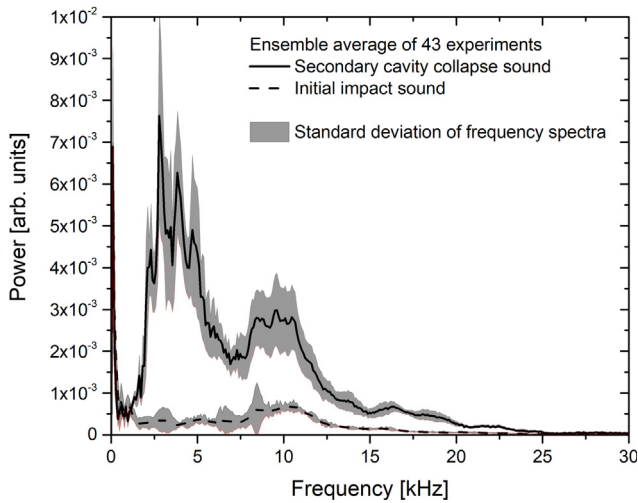


Fig. 7. Frequency spectra for two sound events observed during drop impacts into a deep liquid pool (Fig. 6). Solid line is secondary cavity collapse sound event; dashed line is initial impact sound; and gray bands represent the standard deviation in measured frequency spectra.

pronounced feature occurs when the air cavity, created by the impinging drop, collapses and the Worthington jet is issued. The second excitation lasts for around 10 ms, which is similar to the timescale of jet growth to its maximum vertical extent. While recreating the experiments of Tilger et al. [20,21] in our laboratory; a single drop impacts a

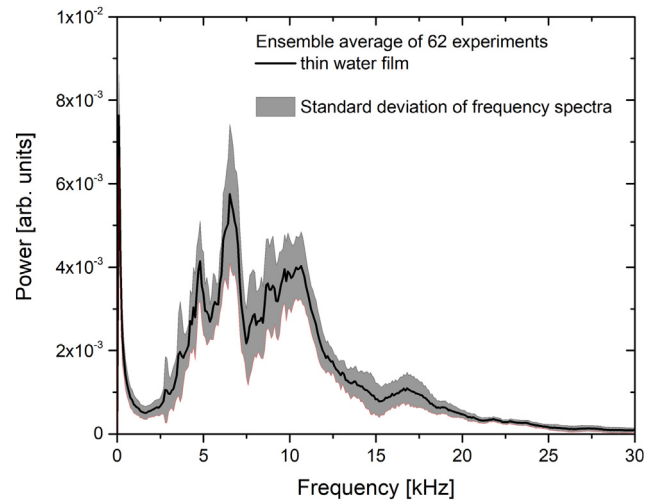


Fig. 9. Frequency spectra for drop impacts onto thin film of water on a smooth aluminum surface (solid line). The gray bands represent the standard deviation in measured frequency spectra.

hydrophobic substrate at fairly low We , spreads into a toroidal volume of fluid, and then jets from the axisymmetric centerline. And in other drop-drop or drop-geometry interactions where jetting was observed, an audible 'tick' exceeding non-jetting cases was often heard.

For each of the deep pool microphone-captured audio signals, a rectangular window fast Fourier transform (FFT) was performed independently on the first feature (impact) and the second feature (cavity collapse), to decompose the frequency content of these two sounds. The initial impact and cavity collapse sounds are well isolated and separated by around 50 ms. Frequency analysis for the impact sound is performed on the first 10 ms of data and for the secondary cavity collapse sound is performed on 20 ms of data following the start of that sound event. The transforms were performed across 501 discrete frequencies, providing a resolution of 96 Hz on the domain of 0–48 kHz. Background noise in each impact frequency spectra was removed by subtraction of a measured background spectra, taken immediately prior to the impact event. Frequency spectra for many impact events (29–62 impact events) were ensemble averaged to obtain the final frequency spectra. As shown in the below figures, the standard deviation in the measured frequency spectra is modest, illustrating the repeatability in the measurements.

As shown in Fig. 7, the initial sound at impact has frequency content centered around 10 kHz (8–12 kHz) and with far less energy (area under curve) than the second feature. The much stronger secondary sound, has additional observed frequency content from 2 to 8 kHz, very similar to the frequencies observed underwater by Medwin et al. [15] due to air bubble oscillations after the impact of 2.5–5 mm drops into pools.

For the impact of a liquid drop onto the wetted aluminum plate, the

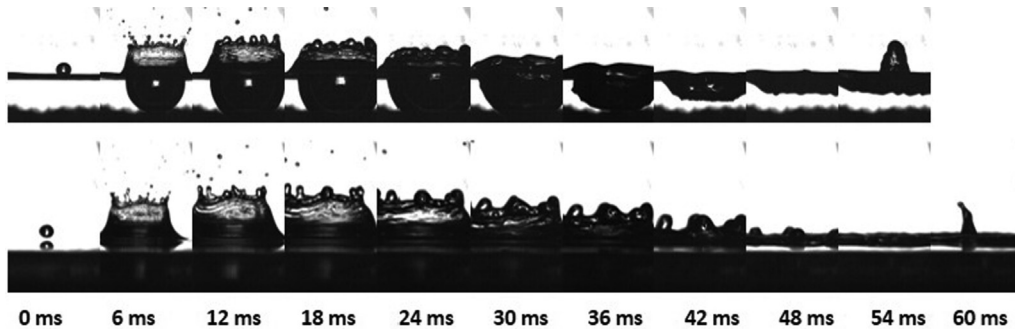


Fig. 8. Photographs comparing the resultant crown splash and air cavity formation for water drop impacts into deep pool (top) and onto thin film of water on aluminum surface (bottom).

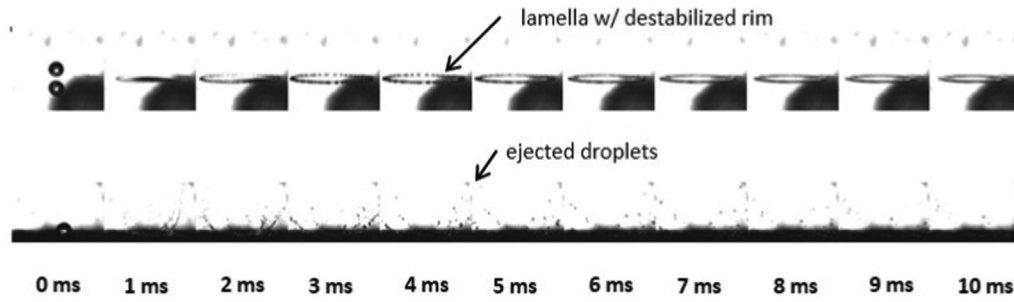


Fig. 10. Photographs capturing the impact of a water drop on dry aluminum (top) and masonry brick surface (bottom).

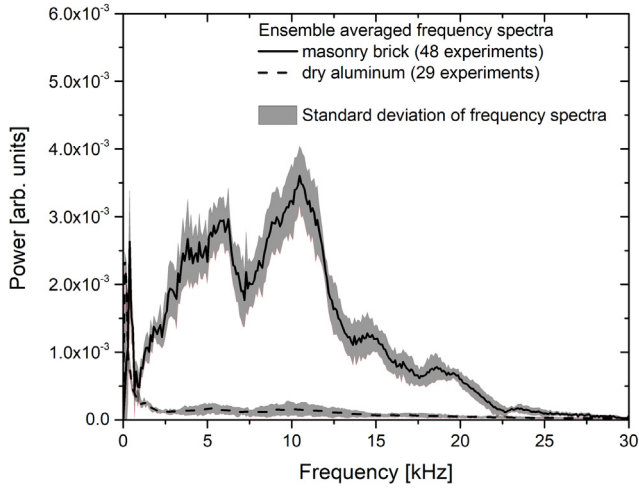


Fig. 11. Frequency spectra for drop impacts onto masonry brick (solid line) and dry smooth aluminum surface (dashed line). The gray bands represent the standard deviation in measured frequency spectra.

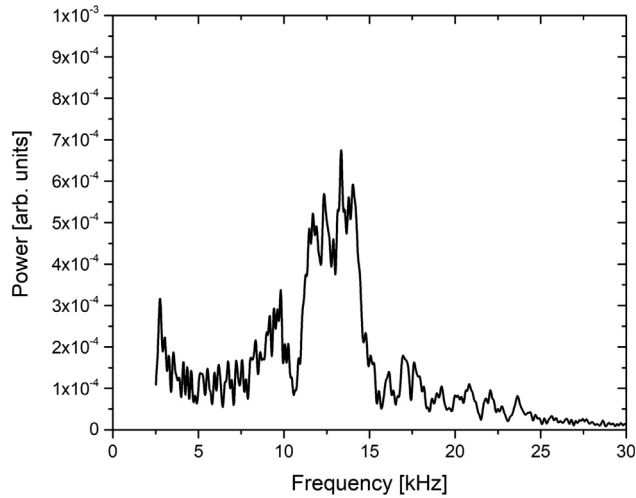


Fig. 12. Frequency spectra for natural rainfall on wet concrete.

fluid mechanic response is similar to the impact into a deep pool, as shown in Fig. 8. The crown splash behavior is also observed with an associated air cavity. However, the presence of the rigid obstacle at the bottom of the film affects the cavity shape and subsequently jet formation [22]. The radiated sound frequency spectra for impact onto the wetted surface are shown in Fig. 9. For the wetted surface, like the pool, the impact produces two sounds. The first, is the sound at impact with a frequency centered again around 10 kHz. In the case of the wetted surface this sound is larger in magnitude. The collapse of the air cavity gives rise to a secondary sound at frequencies from 3 to 8 kHz, with a

peak at somewhat higher frequencies than observed in the deep pool (7 vs 3 kHz), behavior that is thought to arise from the solid surface constraining the air cavity to a smaller volume.

Images of drop impacts on dry aluminum and masonry surfaces are shown in Fig. 10. The frequency spectra derived from the radiated sound of each geometry are displayed in Fig. 11. In the case of the dry aluminum, the drop impact results in a spreading of the liquid into a thin disk with a periodically-wavy lamella. Limited lamella recession is observed due to the surface chemistry of the aluminum (moderately hydrophilic), with no recoil/bounce of impacted volume of fluid. These well-behaved impacts onto substantial, smooth substrates result in fairly weak and relatively broadband (0–15 kHz) sound levels. Prior studies have shown that a small cylindrical air cavity is created from the impact and spreading of a liquid drop on a smooth solid surface [22,25], whose collapse presumably causes the low-intensity sound signature observed here for the smooth dry aluminum surface drop impact case.

For the masonry surface, drop-diameter-scale surface roughness channels the impacting fluid into the valleys between various features. Compared with smooth surfaces, this geometry drastically shifts towards chaotic breakup of the drop with jetting of thin fluid elements and droplet ejection. These behaviors lead to a relatively broadband, but strong, acoustic response with pockets of energy centered around 5 and 10 kHz. In the case of the masonry surface, the radiated sound is clustered within a millisecond of drop impact, due to the rapid fluid dynamics of drop breakup and droplet ejection.

Measurements of the frequency power spectra for natural rainfall (rate of 2 ± 0.5 mm/hour) on the wet concrete are shown in Fig. 12. The frequency spectra was determined by superimposing the Fast Fourier Transforms from sixty 52 ms audio clips. An FFT across 2501 discrete frequencies provided a resolution of 19.2 Hz on the domain of 0–48 kHz. The natural rainfall shows a strong feature from approximately 8–15 kHz that is very similar in shape to the feature assigned to the single drop impact event in the laboratory single-drop studies on masonry (Fig. 11); hence, the frequency spectrum appears to be a simple ensemble of many single drop impact events. The 8–15 kHz feature, arising from air cavities at the liquid drop interface with the porous surface, is slightly shifted to higher frequencies for natural rainfall, presumably due to the mostly smaller size of the natural rain drops, relative to the drops used in the laboratory experiments, forming smaller air cavities that result in higher frequencies. This finding is in concert with the underwater study of Medwin et al. [15] on the dependence of radiated sound frequency on impacting drop size.

Spiel [26] has previously shown that a Helmholtz resonator model can be used to describe sound generation from bursting air bubbles on the surface of a liquid pool. A Helmholtz resonator (Fig. 13, left side) is a spherical echo chamber with an opening for sound waves to travel out and be heard. The resonator geometry is such that only a certain frequency and its harmonics will propagate out, undamped to the listener's ear. The spherical Helmholtz resonator is similar in geometry to the cavity that is formed after impact of a drop into a liquid pool or liquid film, as shown in Fig. 13. Spiel derived the resonant frequency

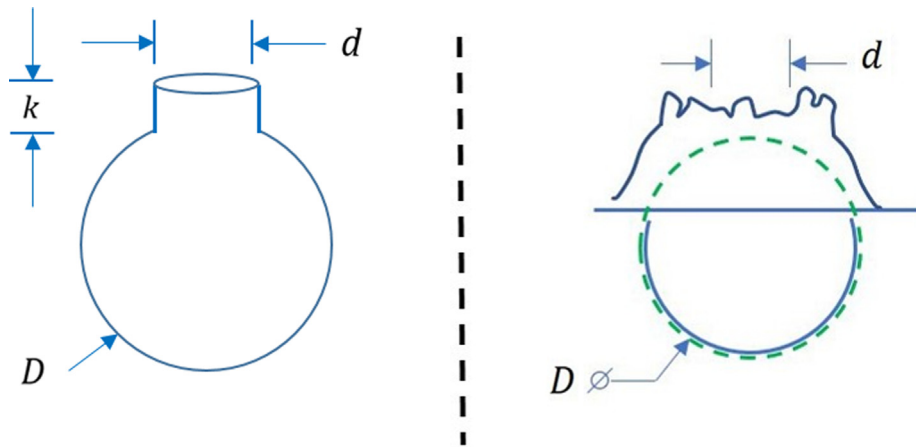


Fig. 13. A schematic of a classic Helmholtz resonator (left) and model for an air cavity which might occur after a drop impact into a pool (right).

formula for this system to be:

$$f = \left(\frac{c}{4\pi} \right) \sqrt{\frac{3\left(\frac{d}{2}\right)}{k\left(\frac{D}{2}\right)^3}}$$

where c is the speed of sound in air, d is the diameter of the opening, k is a factor representing the effective length of the neck, and D is the diameter of the air cavity or sphere in question. Spiel reported k as being a value between $8\pi/3$ and $16\pi/3$ for drops impinging onto liquid pools. Applying this model to laboratory drop impacts into a liquid pool or film results in predicted frequencies from 3 to 7 kHz for the second sound event coinciding with the collapse of the air cavity created by the impinging liquid drop, where the inputs for d and D were estimated from the high-speed images just prior to the cavity collapse to be 6–7.5 mm and 6–10 mm, respectively. This estimate agrees well with the experimental audio frequencies observed for liquid pools and films, 2–8 kHz, especially given that the geometry of the collapsing cavity only qualitatively resembles the necked sphere Helmholtz resonator model. Note, the Helmholtz resonator model does not apply to drop impacts on solid surfaces (masonry or aluminum) which do not yield the air cavities of the same type.

4. Summary

The sound generated by the impact of liquid water drops on four surfaces has been investigated using audio sampling for determining the frequency power spectrum of radiated sound and the acquisition of high-speed images to observe fluid dynamic behavior. Impact studies for water drops 2.1 mm in diameter at a velocity of 5.9 m/s into a deep pool of water, a wetted aluminum surface, a dry aluminum surface, and a dry porous masonry surface were carried out in the laboratory. In these experiments, time-dependent sounds are related to events such as first contact and air cavity collapse observed on synchronized high-speed images. The sound was recorded with significant spectral energy at frequencies up to 20 kHz. In all impact events, there was a concentration of sound energy centered around 10 kHz. In the deep pool and film cases, after the initial impact a subsequent air cavity collapse resulted in significantly more sound energy, but shifted towards lower frequencies (2–8 kHz). A Helmholtz resonator model applied to the collapse of air cavities in water predicts frequencies for these events (3–7 kHz) that are similar to that measured in the laboratory.

Additionally, the sound frequency power spectrum for natural rainfall on a masonry surface with roughness features on the order of the impacting drop diameter was collected. The natural rain drop impacts resulted in a frequency spectrum comparable to that observed in the laboratory for single-drop impacts on masonry, but slightly

upshifted due to the smaller size of the natural rain drops.

CRediT authorship contribution statement

Sean T. Beacham: Data curation, Formal analysis, Investigation, Methodology, Software, Writing - original draft. **Christopher F. Tilger:** Methodology, Software, Visualization, Writing - review & editing. **Matthew A. Oehlschlaeger:** Supervision, Methodology, Conceptualization, Writing - review & editing.

Declaration of Competing Interest

I hereby certify that the authors have no conflicts of interest, financial or otherwise, and this manuscript has not been submitted elsewhere.

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